Rohan Devraj

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Education

Georgia Institute of Technology | Atlanta, GA *Expected Graduation May 2027*
B.S. in Industrial & Systems Engineering, Minor in Computer Science *GPA: 3.84*

- *Concentrations*: Operations Research, AI & Machine Learning
- *Relevant Coursework*: Applied Probability, Optimization, Statistics, Data Input & Manipulation, Discrete Math, Data Structures & Algorithms, Stochastic Systems, Database Systems

Experience

Optum Insights - Analytics (UnitedHealth Group) Jun. 2025 – Aug. 2025
Data Engineering Intern *Minneapolis, MN*

- Developed a Spark-based Python library deployed in Azure Databricks to compute AHRQ Quality & Patient Safety measures from ICD-10 healthcare data, replacing legacy SAS systems and saving $36K annually
- Reduced processing time by 90% (10 hrs to < 1 hr) by optimizing with Pandas UDFs and Databricks
- Built validation pipelines to benchmark Spark outputs against legacy SAS logic and accelerated health data modernization through fast, in-place cloud processing

Space Systems Optimization Group at Georgia Tech Aug. 2024 – Present
Undergraduate Researcher *Atlanta, GA*

- Developed and optimized a Mixed Integer Linear Program (MILP) in Python with Gurobi for vehicle routing, modeling 5,340 mapped locations with elevation-based path costs (advised by Dr. Koki Ho)
- Solved the model with 52,000+ variables and 62,000+ constraints, achieving optimal route plan in under 8 seconds
- Led research on adaptive path planning to improve dynamic decisions under satellite and terrain data uncertainty

Carnegie Mellon Department of Statistics & Data Science Jun. 2024 – Aug. 2024
Summer Undergraduate Research Fellow | stat.cmu.edu/cmsac/sure/2024/showcase/ *Pittsburgh, PA*

- Nationally selected (1 of 18) for a competitive research program in data science applied to healthcare analytics
- Developed regression and machine learning models (linear, lasso, ridge, decision trees) to analyze the impact of adult health behaviors on child mortality and low birthweight at the county level
- Evaluated and compared models using 10-fold cross-validation; selected linear regression for its 5% lower root mean squared error (RMSE), enhancing low birthweight prediction accuracy
- Conducted 10+ analyses on maternal healthcare disparities and applied k-means clustering to identify distinct demographic and behavioral risk groups

Autonomous & Connected Transportation Laboratory at Georgia Tech Jan. 2024 – May. 2024
Undergraduate Research Assistant *Atlanta, GA*

- Formulated a multi-objective traffic assignment model in Python using Pyomo and Gurobi to optimize routing across zone-based transportation networks, balancing travel time, congestion, and safety across diverse zones

Leadership

H. Milton Stewart School of ISyE, Georgia Tech Aug 2025. – Present
Academic Tutor *Atlanta, GA*

- Tutor 70+ students in advanced 3000-level ISyE courses like Optimization, Statistics, and Stochastic Systems

Georgia Tech Alumni Association Angel Network Aug. 2024 – Present
Senior Associate | gtangelnetwork.com *Atlanta, GA*

- Lead Georgia Tech's first angel investment network, sourcing 20+ startups from a 100+ deal pipeline and supporting over $6.5M in connected capital
- Direct meetings with founders and collaborate with investors to craft detailed investment memos, analyzing product, market strategy, financial performance, and growth potential

Skills & Activities

Programming/Tools: Python, SQL, Java, R, Excel, PySpark, Azure Databricks, SAS, Git, Copilot

Libraries & Modeling: Pandas, NumPy, ggplot2, dplyr, Gurobi, Pyomo, GurobiPy

Activites: 180 Degrees Consulting, Aarohi Indian Classical Arts, GT Venture Capital Club, Grand Challenges LLC